SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 **Total Marks:30**

Annexure A

Dry Run Evaluation

Code of Conduct:

I Y SAI GOPAL REDDY (192524083) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

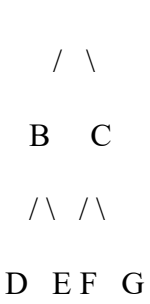
Signature of the student:
Sai gopal reddy
192524083

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:



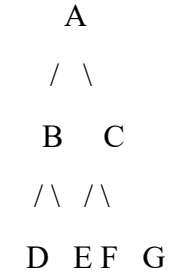
Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration	Queue	Visited Node
1		
2		
3		
4		
5		
6		
7		

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,



Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. **Initial State :**
   1 3 6
   5 0 2
   4 7 8

Goal State :
1 2 3
4 5 6
7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1			
2			
3			
4			
5			

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

LEVEL 4 MODEL ANSWERS

The dry runs below use left-to-right child order for the graph searches. For the 8-puzzle, boards are written as three rows separated by slashes, 0 is the blank, the queue is shown front to rear, and blank moves are generated Up, Down, Left, Right.

Question 1: Breadth-First Search

BFS removes the front node from a FIFO queue, visits it, and adds its children from left to right. The queue column shows the contents after each visit.

Iteration	Queue after expansion	Visited node
1	[B, C]	A
2	[C, D, E]	B
3	[D, E, F, G]	C
4	[E, F, G]	D
5	[F, G]	E
6	[G]	F
7	[]	G

Traversal order: A → B → C → D → E → F → G.

Because the queue is FIFO, every node at one depth is visited before nodes at the next depth. Each graph node is enqueued once, so this trace visits the tree level by level.

Question 2: Depth-First Search

DFS uses a LIFO stack. The stack is shown bottom to top, so its rightmost item is the next node. To visit the left child first, push the right child before the left child.

Iteration	Stack after expansion (bottom → top)	Visited node
1	[C, B]	A
2	[C, E, D]	B
3	[C, E]	D
4	[C]	E
5	[G, F]	C
6	[G]	F
7	[]	G

Traversal order: A → B → D → E → C → F → G.

The stack follows the B branch as deeply as possible before returning to C. This is a preorder, left-first traversal; DFS does not generally guarantee a shortest path.

Question 3: BFS for the 8-Puzzle

The source worksheet's table has five rows, while the goal is removed from the queue on iteration 311. The table below fills those five rows with exact queue contents; the following compact level summary and path complete the dry run without hiding the additional search work. Repeated boards are not enqueued again.

Iteration	Current state	Queue before expansion	Queue after expansion
1	136/502/478	136/502/478	106/532/478; 136/572/408; 136/052/478; 136/520/478
2	106/532/478	106/532/478; 136/572/408; 136/052/478; 136/520/478	136/572/408; 136/052/478; 136/520/478; 016/532/478; 160/532/478
3	136/572/408	136/572/408; 136/052/478; 136/520/478; 016/532/478; 160/532/478	136/052/478; 136/520/478; 016/532/478; 160/532/478; 136/572/048; 136/572/480
4	136/052/478	136/052/478; 136/520/478; 016/532/478; 160/532/478; 136/572/048; 136/572/480	136/520/478; 016/532/478; 160/532/478; 136/572/048; 136/572/480; 036/152/478; 136/452/078
5	136/520/478	136/520/478; 016/532/478; 160/532/478; 136/572/048; 136/572/480; 036/152/478; 136/452/078	016/532/478; 160/532/478; 136/572/048; 136/572/480; 036/152/478; 136/452/078; 130/526/478; 136/528/470

At every later iteration, BFS removes the front state and appends each unseen legal neighbor. It stops when 123/456/780 is removed at depth 8. With this stated move order, 311 states are dequeued through the goal (310 non-goal states are expanded), and 500 unique states are generated/enqueued including the initial state. Of the 500, 499 are new successor states; duplicates are discarded.

Generated-state count by depth

Depth	Unique states enqueued	States dequeued through goal
0	1	1
1	4	4
2	8	8
3	8	8
4	16	16
5	32	32
6	60	60
7	72	72
8	136	110 (goal is last)
9	163	0 (left in queue)
Total	500	311

Shortest solution path

S0 136/502/478 →R S1 136/520/478 →U S2 130/526/478 →L S3 103/526/478 →D S4 123/506/478 →L S5 123/056/478 →D S6 123/456/078 →R S7 123/456/708 →R S8 123/456/780 (goal).

Each legal tile move has cost one. FIFO ordering makes BFS explore states in nondecreasing move count, so the first goal removed from the queue has the fewest possible moves. The path has eight moves, and no shorter solution can exist under this search.